Problem 1

$$A = (1, 0, 1), \quad B = (3, 1, 0), \quad C = (2, -1, 2),$$
$$D_h = (0, 2, h), \qquad h \in \mathbb{R}.$$

1. Compute $\overrightarrow{AB} \times \overrightarrow{AC}$ and the area of triangle ABC.
2. Find a unit normal vector to triangle ABC whose z-component is positive, and find the plane through A, B, C.
3. Express the distance from D_h to this plane as a function of h.
4. Find every h for which tetrahedron $ABCD_h$ has volume 2. Explain briefly why two values occur.

Problem 2

$$L: \quad \mathbf{r} = \langle 1, -1, 2\rangle + t\langle 2, 1, -1\rangle,$$
$$P_\lambda: \quad x + \lambda y + z = 3, \qquad \lambda, t \in \mathbb{R}.$$

1. When $\lambda = 2$, find the point where L meets P_2.
2. Find the value of λ for which L and P_λ have no common point.
3. For the value of λ found in part (2), find the distance from L to P_λ.
4. Find all λ for which the acute angle between P_λ and P_{-1} is 60°.

Problem 3

Consider $S:\ x^2 + 4y^2 - z^2 - 2x + 8y = 4.$

1. Describe the traces in $z = 0$, $x = 1$, and $y = -1$. Use them to sketch S and identify the surface.
2. Find all c for which S intersects the plane $y = c$ in two straight lines. Find the equations of these lines.
3. For $z = k$, find the cross-section area $A(k)$. Find all k such that $A(k) = 13\pi/2$.
4. Find an equation for the surface T of points whose distance to the plane $z = -1$ is twice their distance to $(1, -1, 2)$, and identify T.